JalKavach: Technical Architecture & Implementation Guide

Supplementary Guide

Hyperlocal Flood & Cyclone Early Warning System

Event: Odisha Seva First Innovation Challenge (OSFIC)
Team: IDEA | **Problem Statement:** AI-Based Hyperlocal Disaster Intelligence System
Target Geography: Ganjam District, Odisha

1. Executive Summary & Problem Context

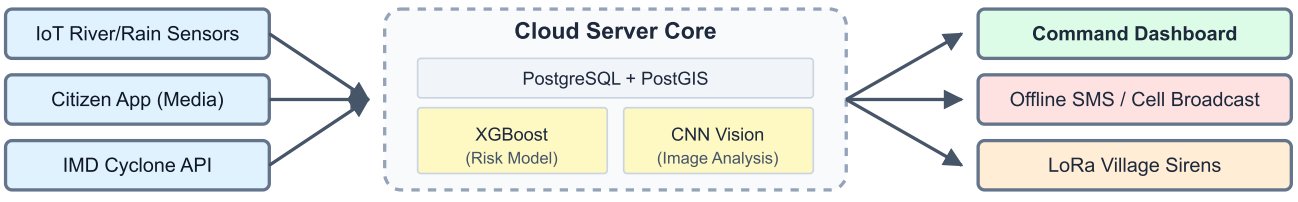
Coastal and riverine regions in Ganjam face severe threats from cyclonic storms, storm surges, and localized flooding. Current early warning systems suffer from two major flaws: they lack hyperlocal, ward-level precision, and they rely exclusively on top-down meteorological data without integrating ground-level realities.

JalKavach solves this by introducing a multimodal AI-based Hyperlocal Disaster Intelligence System. By fusing quantitative IoT sensor data (river water levels, rainfall) and satellite imagery with qualitative ground-truth data (citizen-reported images and text), JalKavach generates a dynamic, real-time Village Risk Score (VRS). This score drives an offline-first Command Dashboard that provides actionable risk alerts and safe evacuation routing for emergency response forces.

2. High-Level System Architecture

JalKavach operates across four integrated layers to ensure data flows seamlessly from the ground to the Command Dashboard, even during network disruptions.

- **Layer 1: Data Ingestion:** Collects automated telemetry from solar-powered IoT Sentinel Nodes and crowdsourced qualitative data from the JalKavach Citizen App.
- **Layer 2: Edge Communication:** Uses LoRa mesh networking to bypass broken cellular networks.
- **Layer 3: Cloud AI Processing:** A FastAPI/PostgreSQL backend processes incoming streams, applying machine learning models to calculate risk scores.
- **Layer 4: Command Dissemination:** A Zero-Dependency PWA visualizes data for officials and triggers offline SMS alerts.



3. The Multimodal AI & Machine Learning Pipeline

JalKavach utilizes a fused AI architecture to handle both tabular numerical data and unstructured media, ensuring a highly accurate vulnerability assessment.

- **Quantitative Processing:** Time-series data from river sensors, 24-hour rainfall gauges, and storm surge models are fed into an XGBoost model. This predicts the baseline probability of localized inundation based on historical flood thresholds in Ganjam.
- **Qualitative Processing:** Citizen-submitted texts are parsed using Natural Language Processing (NLP) to extract hazard keywords. Citizen photos undergo Computer Vision analysis (CNNs) to estimate the percentage of water/mud in the frame and determine localized water depth.
- **Data Fusion (The VRS Algorithm):** The probabilities from both the quantitative and qualitative pipelines are weighted and combined to generate the final Village Risk Score (0-100). A score above 75 triggers a "Red/Evacuate" status.



4. Edge Computing & Offline-First Dissemination

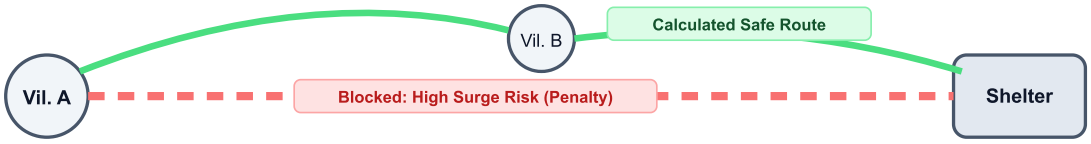
During severe cyclones, cellular infrastructure is often the first to fail. JalKavach is engineered to operate resiliently in zero-connectivity environments across the Ganjam coast.

- **Offline Web Architecture:** The Command Dashboard is built as a Progressive Web App (PWA). It uses Service Workers to securely cache map tiles, UI assets, and shelter coordinates so the dashboard functions seamlessly without an active internet connection.
- **SOS Queuing System:** When citizens request rescue via the offline app, the system generates a native SMS payload containing their exact GPS coordinates. This bypasses the need for 4G/5G data, utilizing the cellular network's underlying control channel.

5. Smart Evacuation & GIS Routing Logic

Unlike standard map routing (which defaults to the shortest path), JalKavach dynamically alters evacuation routes based on live flood data using an adapted Dijkstra's pathfinding algorithm.

Villages and Cyclone Shelters act as "nodes," and connecting roads act as "edges." Under normal conditions, the algorithm routes citizens to the nearest shelter based on physical distance. However, if a road passes through an area with a VRS > 55 (Orange/Red alert indicating storm surge or flooding), the system applies a heavy mathematical penalty to that road's edge weight. This forces the algorithm to instantly recalculate and display a safer, higher-ground alternative route.



6. Hardware Feasibility & Bill of Materials (BoM)

To ensure scalability across the Ganjam district coastline and riverbanks, JalKavach relies on low-cost, ruggedized IoT Sentinel Nodes. Below is the estimated Bill of Materials for a single riverbank deployment unit:

Component	Specification / Purpose	Estimated Cost (INR)
Microcontroller & Comm.	ESP32 with LoRa SX1262 Transceiver (Edge processing)	₹1,200
Sensory Array	Ultrasonic Water Level Sensor + Tipping Bucket Rain Gauge	₹2,300
Power System	5W Solar Panel + LiFePO4 Battery Pack (Monsoon resilient)	₹1,500
Housing	IP67 Weatherproof Enclosure & Mounting brackets	₹500
	Total Estimated Cost per Node:	**~₹5,500**

At this accessible price point, the District Administration can deploy dense, highly redundant sensor networks along the Rushikulya river basin for a fraction of the cost of traditional meteorological stations.

7. Command Dashboard Functional Flow

The JalKavach Progressive Web App (PWA) follows a streamlined operational flow to ensure users and administrators can access critical data quickly. The flow emphasizes offline accessibility, allowing core tasks—such as viewing cached routes, analyzing local photos, and queueing SOS messages—to complete locally when network connectivity is lost.

